

1. You have been tasked with creating operations for the Hospital Database. The database has 3 tables — Patients, Doctors and Admissions. The schema and sample data are given below.

PatientID	FirstName	LastName	DOB	BloodType
1	Rebecca	Adler	1975-09-15	O+
2	Ali	Mohammed	2010-02-01	A-
3	John	Doe	1980-01-01	AB+
4	Mark	Boston	1992-11-30	B-
5	Esteban	Hernandez	1968-06-10	O-

DoctorID	FirstName	LastName	Specialty	HireDate
701	Gregory	House	Diagnostics	1995-04-11
702	Lisa	Cuddy	Endocrinology	1990-07-01
703	James	Wilson	Oncology	1992-05-20
704	Allison	Cameron	Immunology	2004-11-16
705	Robert	Chase	Intensive Care	2004-11-16
706	Eric	Foreman	Neurology	2004-11-16

AdmissionID	PatientID	DoctorID	AdmissionDate	Diagnosis
1001	1	701	2024-10-20	Vasculitis
1002	2	704	2024-10-21	Infection
1003	3	706	2024-10-22	Amnesia
1004	4	705	2024-10-23	Cardiac Arrest
1005	5	703	2024-11-05	Leukemia
1006	1	701	2024-11-06	Follow-up

Answer the following questions based on the tables above.

- (a) (1 point) Write a SQL query to find all blood types that are shared by more than one patient. The query should return the BloodType and the PatientCount in descending order by PatientCount.

0.5
SELECT BloodType, COUNT(*) AS PatientCount
FROM Patients
GROUP BY BloodType
ORDER BY PatientCount DESC;

- (b) (1 point) Write a SQL query to find all patients who meet either of these conditions:
- Their LastName is greater than five letters long.
 - Their BloodType ends with 'Negative' (e.g., 'O-', 'AB-').

0.5
SELECT * FROM Patients
WHERE BloodType LIKE "%-"
UNION
SELECT * FROM Patients
WHERE len(LastName) > 5;

- (c) (3 points) The diagnosis for 'Rebecca Adler' was determined to be 'Cluster Headache'. Write the SQL query to update the most recent admission for Rebecca Adler by AdmissionDate. (Note: You have to write a general query that updates the latest record in Admissions, you can not use the AdmissionID directly)

UPDATE Admissions

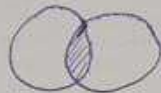
SET Diagnosis = "Cluster Headache"

WHERE AdmissionID = (SELECT AdmissionID FROM Admissions
WHERE PatientID = (SELECT PatientID FROM Patients
WHERE FirstName = "Rebecca" AND
LASTName = "Adler")
AND AdmissionDate = (SELECT AdmissionDate FROM Admissions
WHERE PatientID = (SELECT PatientID FROM Patients
WHERE FirstName = "Rebecca" AND
LASTName = "Adler")
ORDER BY AdmissionDate DESC
LIMIT 1);

- (d) (6 points) This part has two sub-parts.

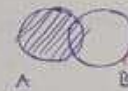
A) (2 points) List and briefly describe the four main types of SQL JOIN operations.

(i) Inner JOIN



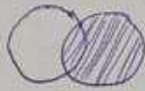
Only common
tuples to
left and right
tables show up in
Joined Table.

(ii) Left JOIN



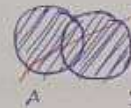
Only left table tuples
along with intersection
show up in the
Joined Table

(iii) RIGHT JOIN



Only right table
tuples along with
intersection show
up in Joined table

(iv) Outer JOIN



All tuples show up in Joined
Table (Left Table &
Right table)

- B) (4 points) Write a single SQL query to list every doctor and the total number of admissions they have handled. (Note: You must use a JOIN operation; conditional SQL queries are not allowed).

- Your result should show the doctor's ID, First and Last Name, and their admission count.
- If a doctor has handled zero admissions, they must still appear in the list with a count of 0.
- Sort the results by the admission count, from highest to lowest.

SELECT Doctor's ID, FirstName, LastName, Count(AdmissionID) AS AdmissionCount
FROM Doctors

OUTER JOIN Admissions on Admissions, DoctorID = Doctors.DoctorID

OUTER JOIN Patients on Patients, PatientID = Patients.PatientID

GROUP BY Doctors.DoctorID

ORDER BY AdmissionCount DESC

the 'Cluster Headache' - stor by Ad-

2. Consider the following single table named StudentEnrollment, which stores all information about students, the courses they are enrolled in, and the professors who teach those courses.

StudentID	StudentName	CourseID	CourseName	ProfName	ProfOffice
S1001	Alice Chen	CS101	Intro to CS	Dr. Elara	D-210
S1002	Bob Davis	CS101	Intro to CS	Dr. Elara	D-210
S1001	Alice Chen	MTH200	Calculus II	Dr. Khan	F-105
S1003	Charlie Fox	MTH200	Calculus II	Dr. Khan	F-105
S1004	Dana White	PHY301	Adv. Physics	Dr. Boro	D-210

Using only the StudentEnrollment table above, answer if an anomaly is present here (INSERT/UPDATE/DELETE). Assume that the data is complete and that there is no other entries except these. Justify your answer clearly.

- (a) (2 points) Another student, 'Joan Smith' has enrolled in the course MTH200 ('Calculus II') course.

INSERT Anomaly, While Inserting Student Details, ProfDetails and CourseDetails should also be Inserted.

- (b) (2 points) The 'D' building wing is being renovated. All professors in office D-210 must move.

- Dr. Elara is moving to a new office, E-301.
- Dr. Boro is moving to a new office, E-302.

(1) UPDATE StudentEnrollment
SET ProfOffice = 'E-301'
WHERE ProfName = 'Dr. Elara'

(2) UPDATE StudentEnrollment
SET ProfOffice = 'E-302'
WHERE ProfName = 'Dr. Boro'

No Anomaly but redundant updates are being done.

Instead of updating one-row, all rows are updated.

Normalised Tables form

- (c) (2 points) Student 'Dana White' (S1004) decides to drop her 'Adv. Physics' (PHY301) course.

DELETE Anomaly

If the row is dropped, We lose Dr. Boro's Information as well.

anomaly?

1/2

3. (2 points) A new database analyst is trying to find the total sales for each EmployeeID from the Sales table, but only for sales where the SaleAmount was over \$100. (Note: Assume a Sales table with at least EmployeeID and SaleAmount columns exists). The analyst write two queries.

Query A:

```
SELECT EmployeeID, SUM(SaleAmount)
FROM Sales
WHERE SaleAmount > 100
GROUP BY EmployeeID;
```

Query B:

```
SELECT EmployeeID, SUM(SaleAmount)
FROM Sales
GROUP BY EmployeeID
HAVING SaleAmount > 100;
```

Assuming 1
Employee can
have more than
1 sale amount
entry.

Determine which query, if any, correctly fulfills the analyst's request, and justify your reasoning. If any query fails, explain what was the reason for it doing so.

Query B succeeds, A fails

Also E is correct, meaning SaleAmount > 100, which is not considered before grouping by, we eliminate the possibility for one employee having

sales → Where $<100 + <100 = >100$, e.g. $70 + 80 = 150$
 $<100 <100$

But A → This case won't be ignored if first employees are grouped by ID and then SaleAmount are seen.

For ex. Person 5 → Having SaleAmount > 100 as only 2nd entry, will be considered.

4. (3 points) You are building SQL queries for a database but the legacy system does not seem to have the FULL OUTER JOIN operation built in along with other JOIN operations. Quick, use the basic SQL commands to compute the FULL OUTER JOIN of two tables joined on an attribute.

SELECT * FROM Table 1

LEFT JOIN Table 2 ON Table 1.Attr 1 = Table 2.Attr 1

UNION

SELECT * FROM Table 1

RIGHT JOIN Table 2 ON Table 1.Attr 1 = Table 2.Attr 1.



5. (6 points) You are given a relation $R(A, B, C, D)$ and the following set of functional dependencies (FDs):

- $A \rightarrow B$
- $BC \rightarrow D$
- $D \rightarrow A$

(a) What are all the candidate keys for this relation? [1 point]

$$\begin{aligned} A^+ &= \{A, B\} \\ \{B, C\}^+ &= \{B, C, D, A\} \\ D^+ &= \{D, A, B\} \end{aligned} \quad \left. \vphantom{\begin{aligned} A^+ &= \{A, B\} \\ \{B, C\}^+ &= \{B, C, D, A\} \\ D^+ &= \{D, A, B\} \end{aligned}} \right\} \text{Closures}$$

AC BC DC

$\therefore BC \rightarrow$ Candidate Key 0-5

Super Keys: ABC, BCD, AC

Minimal Super Key
(i.e. Candidate Key)

(b) Is this relation in 3NF? Explain your reasoning. [3 point]

$$BC \rightarrow D$$

(Prime) (Non-prime)

$$D \rightarrow A$$

(Non-prime) (Non-prime)

$$A \rightarrow B$$

(Non-prime) (Part of Prime)

Violates Transitivity Property, which shouldn't exist in 3NF

In 3-NF:

(i) Already in 2NF

(ii) $X \rightarrow Y \quad \forall \text{ FDs,}$

X : Super Key (OR)

Y : Prime Attribute

$\therefore$ 2 FDs violate, $D \rightarrow A$ and $A \rightarrow B$

(c) Is this relation in BCNF? Explain your reasoning. [2 point]

If Relation not in 3NF $\Rightarrow$ Relation not in BCNF

Also for BCNF:

(a) Relation should be in 3NF

(b) $X \rightarrow Y \quad \forall \text{ Non-Trivial FDs,}$

X : Super Key

$$BC \rightarrow D$$

(Super Key)

$$D \rightarrow A$$

$$A \rightarrow B$$

Not in BCNF



6. (12 points) The Department of Computer Science at a university maintains a database to keep track of the courses it offers each semester. The current database schema stores details such as the course name, instructors, departments, and textbooks used. However, the database was designed by a new intern, and the data has not been properly normalized. As a result, several issues have emerged. Note: Both of the following tables are part of the same database schema.

course_id	course_name	instructor_id	instructor_name	instructor_phone
C101	Database Systems	I11, I14	Dr. Mehta, Dr. Singh	9876543210, 9876543215
C102	Operating Systems	I12	Dr. Rao	9876543211
C103	Data Structures	I13	Dr. Patel	9876543212
C104	Linear Algebra	I21	Dr. Sharma	9876543213
C105	Calculus I	I21	Dr. Sharma	9876543213
C106	Database Systems Lab	I11	Dr. Mehta	9876543210
C107	Algorithms	I13, I15	Dr. Patel, Dr. Verma	9876543212, 9876543216
C108	Probability	I22	Dr. Banerjee	9876543214
C109	Machine Learning	I11	Dr. Mehta	9876543210
C110	Numerical Methods	I22	Dr. Banerjee	9876543214

department_id	department_name	textbook_title	textbook_author
D10	Computer Science	Database System Concepts, Fundamentals of DB	Silberschatz
D10	Computer Science	Modern Operating Systems	Tanenbaum
D10	Computer Science	Data Structures and Algorithms	Weiss
D20	Mathematics	Linear Algebra Done Right	Axler
D20	Mathematics	Calculus Early Transcendentals, Calculus Made Easy	Stewart
D10	Computer Science	Database System Concepts	Silberschatz
D10	Computer Science	Introduction to Algorithms	Cormen
D20	Mathematics	Introduction to Probability Models	Ross
D10	Computer Science	Pattern Recognition and Machine Learning	Bishop
D20	Mathematics	Numerical Analysis	Burden

- (a) List all Functional Dependencies (FDs) that hold in the given schema, based on the data shown. [2 points]
- (b) Normalize the schema to 1NF (3 points), 2NF (4 points) and 3NF (3 points). At each stage clearly state the reason for normalization and the new schema(s) obtained. If a relation is already in a normal form, explicitly mention it and explain why. [10 points]

(a) $course_id \rightarrow course_name$

$instructor_id \rightarrow \{instructor_name, instructor_phone\}$

$department_id \rightarrow department_name$

$course_name \rightarrow textbook_title$

$course_id \rightarrow dept_id$
 $textbook_title \rightarrow textbook_author$

(b) 1NF:

R_1	R_2
C D E	A B C F G H I

1NF not mentioning how multivalued are handled (-1)

Assuming:

A: $course_id$

B: $course_name$

C: $instructor_id$

D: $instructor_name$

E: $instructor_phone$

F: $department_id$

G: $department_name$

H: $textbook_title$

I: $textbook_author$

R_1	R_2
C D E	A B C F G H I

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For 2NF:

Through 1NF, we obtained:

$C \rightarrow \{DE\}$
 $D \rightarrow \{CE\}$
 $E \rightarrow \{C\}$

R_1 already in 2NF

$A \rightarrow B$ ✓

(1) Already in 1NF

(2) Subset of CK $\rightarrow$ Non-prime
is not allowed

i.e. No Partial Dependency

- 2 since no keys are mentioned
so, how can you involve dependencies?

For 3NF:

Course_ID	Course_Name	Instructor_ID

(i) Already in 2NF

(2) $X \rightarrow Y$

X: Super key
(OK)

Y: Prime Attribute

Instructor_ID	Instructor_Name	Instructor_Phone

Department_ID	Department_Name	Course_ID

Textbook_Author	Textbook_Author	Course_ID

- 1.5 since no keys are mentioned

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